McMaster University | Science

Chemistry Chemical Biology
A.N. Bourns Building, Room 156
1280 Main Street West,
Hamilton, ON L8S 4L8

Dear BIMR Director,

I am writing to express my strong support for the grant proposal titled *"Optimizing Conductivity via Genetic Algorithms Using Simulated Doping for Solid-State Batteries"*, submitted by Cameron Gurwell from Dr. Gillian R. Goward's group and my M.Sc. student, Uriel Garcilazo Cruz.

Cameron and Uriel have been exploring the use of recently released foundational force field models to accelerate the computational simulation of lithium battery compounds. Given the speed and accuracy of these models, coupling them with search algorithms—such as genetic algorithms—could significantly enhance the exploration of chemical space to identify compounds with optimal properties.

While my group has an active research allocation with the Digital Alliance, high demand for GPUs and long wait times makes it challenging to meet the real-time computational needs of this project. The requested funds will be used to acquire a dedicated GPU workstation capable of efficiently performing geometry optimizations and other simulation pipelines for multiple compounds. This workstation will complement existing computational resources, ensuring that Cameron and Uriel can not only run simulations but also retrain foundational models if needed. If necessary, I am willing to contribute additional funds (~2,000.00 CAD) to support the purchase of a more powerful workstation.

Thank you for considering Cameron and Uriel's application. I look forward to seeing the impact of their research.

Sincerely,

Rodrigo A. Vargas-Hernández

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